

Sample End Semester Question Paper - 2026

Subject: Data Structures | Code: CS301 | Semester: 3

Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. State the base case and the recursive case in recursion.
2. What is the time complexity of binary search?
3. State the worst-case time complexity of quicksort.
4. What is a balanced binary tree? Name its balance condition.
5. Define data structure. Give two examples of linear data structures.
6. State the time complexity of linear search in an array.
7. What is a stack? Name its two basic operations.
8. Convert the infix expression $A + B * C$ into postfix form.
9. State one application of a queue.
10. What is a linked list? State one advantage over an array.

PART B - Long questions (10 x 8 = 80 Marks)

1. Explain hashing using open addressing and separate chaining with an example.
2. Explain stack operations push, pop and peek with algorithms. Illustrate with an example.
3. Explain infix to postfix conversion with a complete algorithm and a worked example.
4. Explain insertion and deletion in a singly linked list at the beginning, middle and end with diagrams.
5. Explain merge sort with a step-by-step diagram and analyse its time complexity.
6. Explain Binary Search Trees: insertion, deletion and search algorithms.
7. Explain AVL tree rotations LL, RR, LR and RL with examples.
8. Explain circular queue with insertion and deletion algorithms. Discuss applications.
9. Explain graph representations (adjacency matrix and adjacency list). Write BFS and DFS algorithms.
10. Compare arrays and linked lists. Discuss insertion complexity in each with examples.